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(12) **United States Design Patent** (10) **Patent No.: US D1,093,608 S**
Zamarripa et al. (45) **Date of Patent: ** Sep. 16, 2025**

(54) **FIDUCIAL MARKER WITH A FLAT TOP
FOR COMPUTER-ASSISTED SURGERY**

2090/3991; A61B 2090/3916; A61B
34/20

See application file for complete search history.

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(56) **References Cited**

U.S. PATENT DOCUMENTS

5,730,130 A * 3/1998 Fitzpatrick A61B 90/39
600/407
7,720,522 B2 * 5/2010 Solar A61B 90/39
600/407
8,386,022 B2 * 2/2013 Jutras A61B 34/20
600/407

(Continued)

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(57) **CLAIM**

The ornamental design for a fiducial marker with a flat top
for computer-assisted surgery, as shown and described.

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DESCRIPTION

FIG. 1 is a perspective view of a fiducial marker with a flat
top for computer-assisted surgery showing the new design;
FIG. 2 is a first side view thereof;
FIG. 3 is a second side view thereof;
FIG. 4 is a front view thereof;
FIG. 5 is a back view thereof;
FIG. 6 is a top view thereof;
FIG. 7 is bottom view thereof; and,
FIG. 8 is a perspective view of an example operational use.
The broken lines in the drawings are included for the
purpose of illustrating portions of the fiducial marker with a
flat top for computer-assisted surgery that form no part of the
claimed design. The additional broken lines included in FIG.
8 are for showing environmental structure that forms no part
of the claimed design. The dot-dash line shown in FIG. 7 is
included for defining a boundary between the claimed and
unclaimed portions of the fiducial marker with a flat top for

(Continued)

(**) Term: **15 Years**

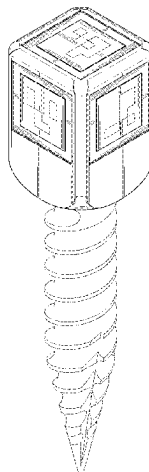
(21) Appl. No.: **29/847,931**

(22) Filed: **Jul. 28, 2022**

(51) **LOC (15) Cl.** **24-03**

(52) **U.S. Cl.**
USPC **D24/155**

(58) **Field of Classification Search**
USPC D24/155, 144, 145, 146, 147, 171
CPC A61B 90/39; A61B 2090/3983; A61B
2090/3937; A61B 2090/3995; A61B



computer-assisted surgery, the dot-dash line itself forming no part of the claimed design.

1 Claim, 5 Drawing Sheets

(56)

References Cited

U.S. PATENT DOCUMENTS

9,008,755 B2 * 4/2015 Kim A61B 90/39
600/417
D844,421 S * 4/2019 McClure F16B 19/109
D8/382
10,925,493 B2 * 2/2021 Forsyth A61B 5/0071
11,033,353 B2 * 6/2021 Roger A61B 90/39
D1,003,702 S * 11/2023 Denham A61B 17/0401
D24/145
D1,039,693 S * 8/2024 Xu D24/146

D1,041,656 S * 9/2024 Zamarripa D24/155
2007/0093709 A1 * 4/2007 Abernathie A61B 90/39
600/407
2007/0100325 A1 * 5/2007 Jutras A61B 90/39
606/1
2010/0192961 A1 * 8/2010 Amiot A61B 90/39
128/882
2011/0071431 A1 * 3/2011 Speeg A61B 90/39
600/567
2014/0276005 A1 * 9/2014 Forsyth G01B 11/16
600/424
2017/0231715 A1 * 8/2017 Roger A61B 34/20
600/424
2018/0235715 A1 * 8/2018 Amiot A61B 34/20
2022/0183762 A1 * 6/2022 Nikou A61B 90/30
2023/0134629 A1 * 5/2023 Amiot A61B 34/20
600/424
2023/0149116 A1 * 5/2023 Stauffer A61B 90/39
606/102

* cited by examiner

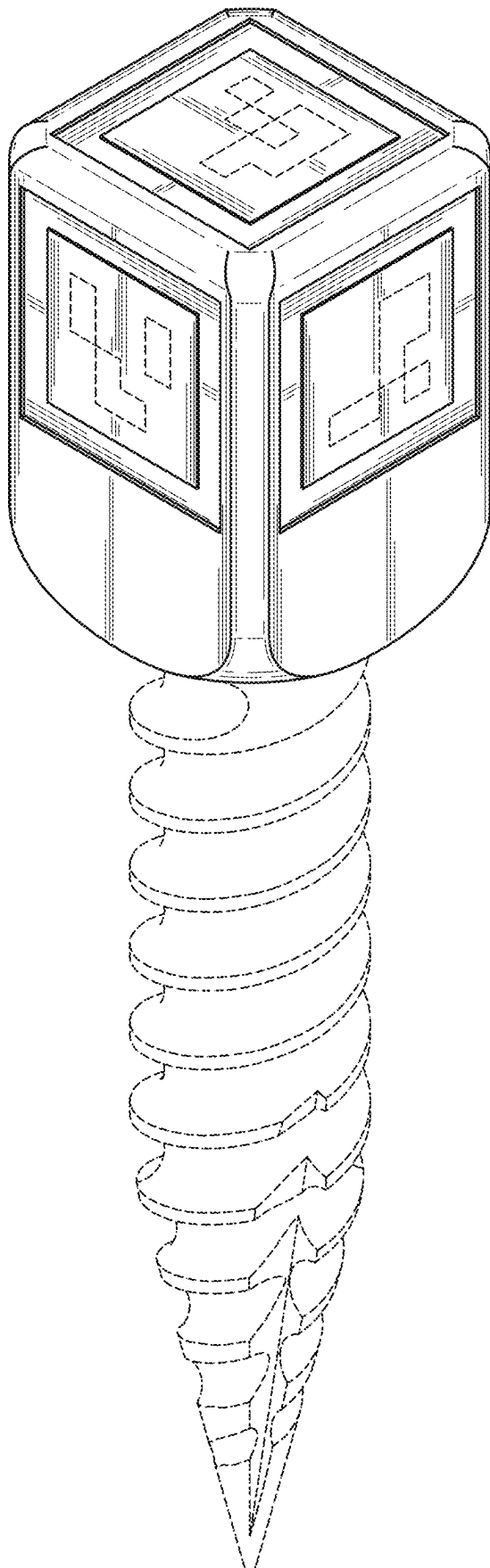


FIG. 1

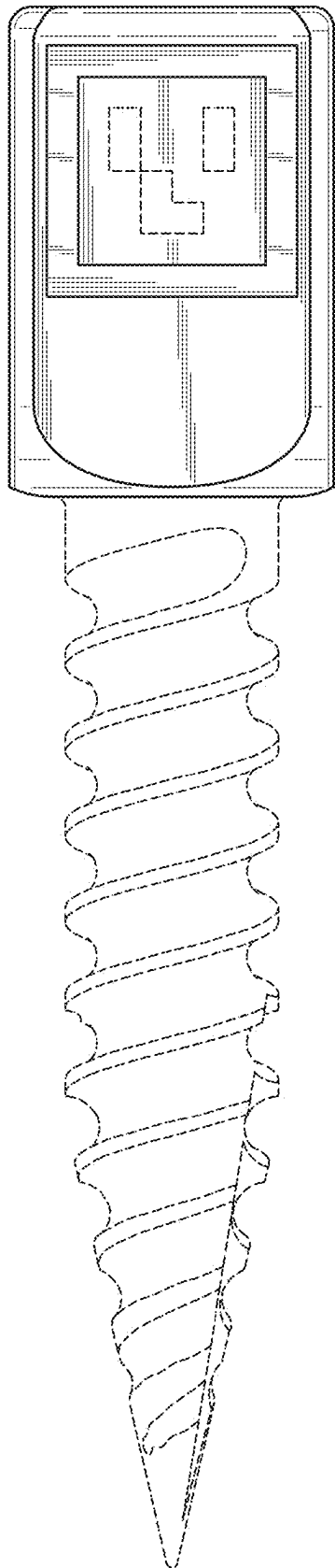


FIG. 2

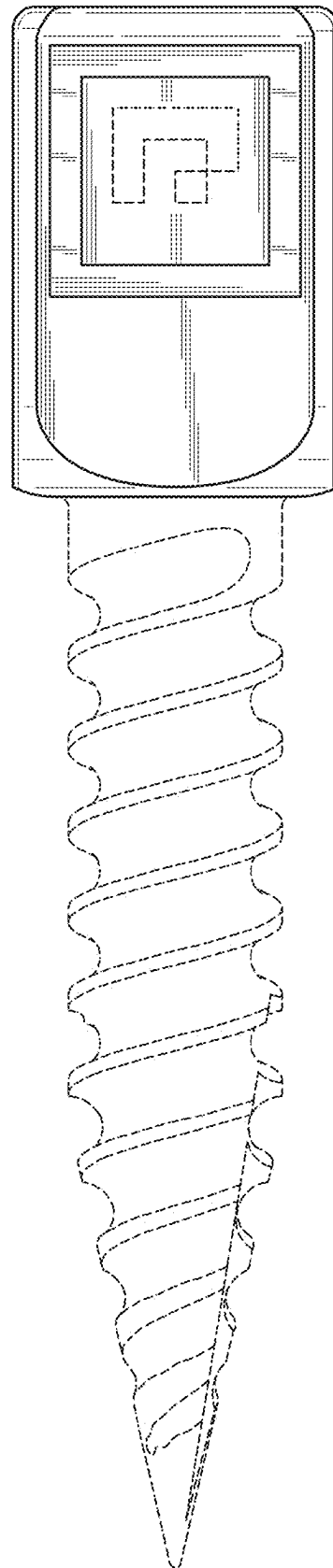


FIG. 3

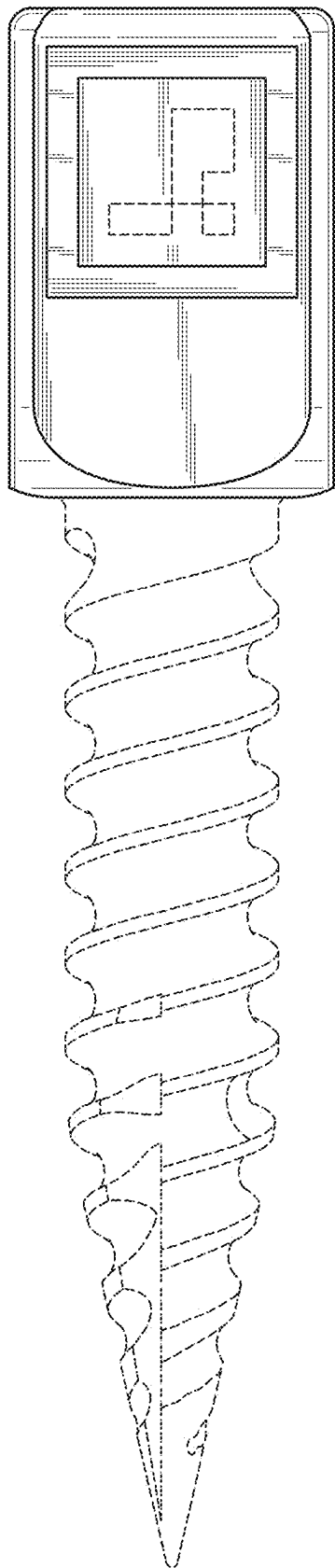


FIG. 4

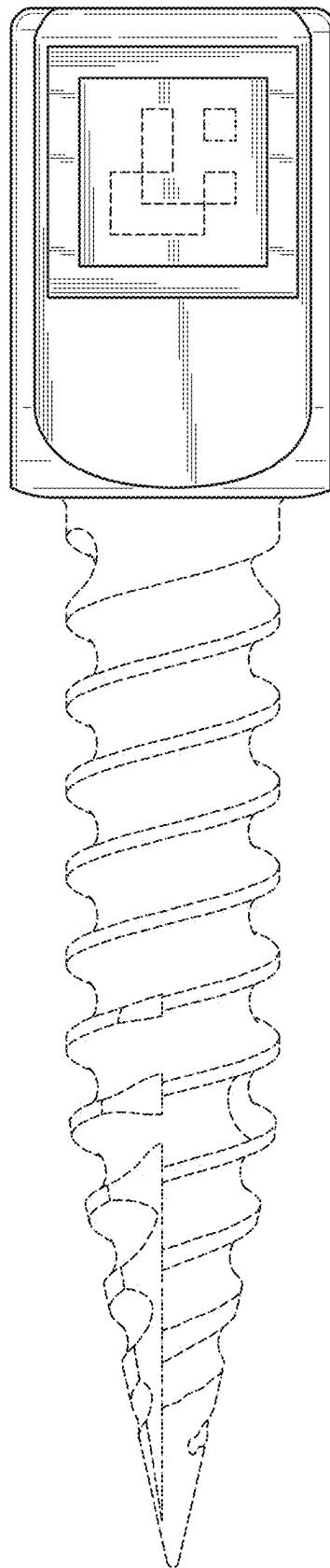


FIG. 5

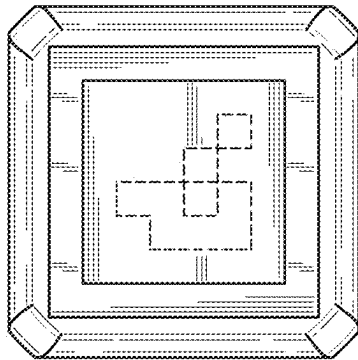


FIG. 6

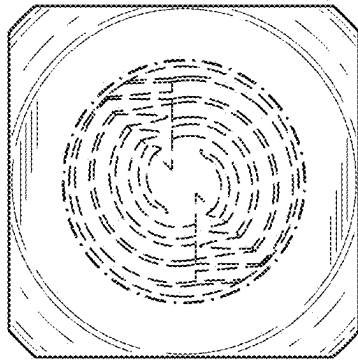


FIG. 7

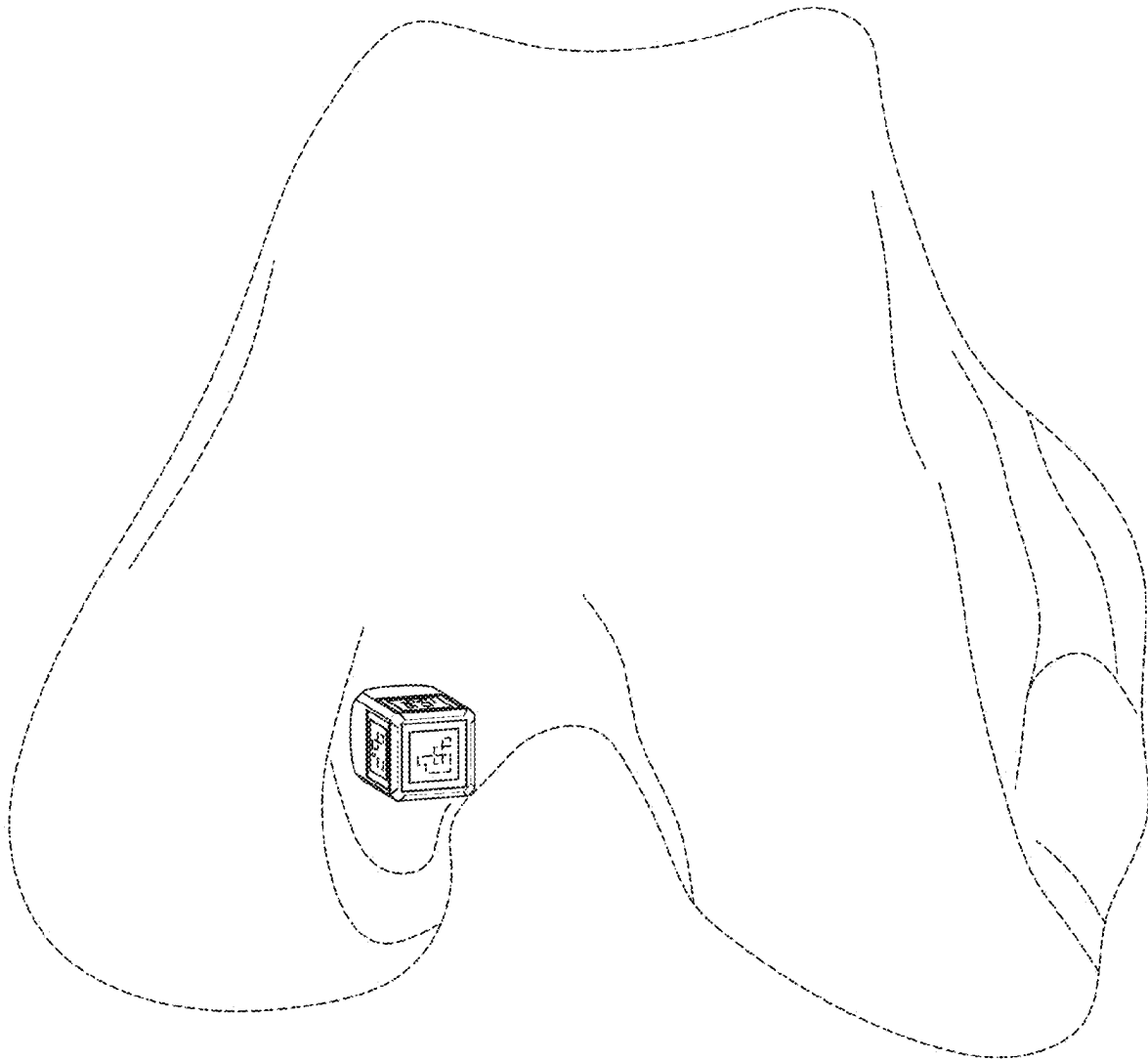


FIG. 8